

TAHLEQUAH, OK, USA

WMO: 722199

Lat: 35.929N

Lon: 95.004W

Elev: 873

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53956

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	17.7	3.3	6.7	18.3	9.0	9.0	22.2	23.6	46.2	20.7	45.7	3.8	330	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	99.4	77.8	96.8	77.2	92.9	76.2	80.4	93.1	79.1	92.1	77.9	90.5	6.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	146.7	87.1	75.3	137.0	84.5	74.5	133.4	83.8	45.0	94.8	43.3	92.2	41.8	90.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.7	17.5	15.7	DB	3.3	103.0	6.5	4.1	-1.4	106.0	-5.2	108.4	-8.9	110.7	-13.6	113.7	
			WB	2.6	81.7	6.4	1.9	-2.0	83.1	-5.8	84.2	-9.4	85.3	-14.0	86.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	37.4	41.2	51.0	59.0	67.4	76.4	80.4	79.7	72.2	59.6	48.8	39.5
	DBStd	17.42	10.29	11.13	10.39	8.71	7.92	4.91	5.15	5.72	6.75	9.29	9.64	9.63
	HDD50	1340	407	288	115	22	3	0	0	0	0	22	135	348
	HDD65	3773	855	668	448	216	69	1	0	0	18	218	490	790
	CDD50	4803	18	40	144	291	543	793	942	922	666	321	100	23
	CDD65	1761	0	1	12	36	144	344	478	457	233	52	4	0
	CDH74	18883	2	8	95	350	1350	3592	5575	5073	2304	499	35	0
	CDH80	8881	0	1	11	63	402	1582	2996	2699	995	129	3	0
Wind	WSAvg	5.4	5.7	6.4	7.1	7.3	5.7	4.8	3.9	3.9	4.0	4.7	5.7	5.5
Precipitation	PrecAvg	47.2	2.3	2.4	3.9	4.3	5.7	5.0	3.7	3.5	4.8	4.2	4.0	3.0
	PrecMax	73.5	6.7	7.6	9.7	8.3	12.1	14.2	13.4	9.6	12.7	12.4	11.3	7.5
	PrecMin	25.7	0.1	0.2	0.6	0.2	1.6	0.9	0.1	0.0	0.3	0.1	0.4	0.3
	PrecStd	9.8	1.7	1.5	2.1	1.9	2.5	2.8	3.1	2.3	3.1	3.1	2.6	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	73.3	81.4	87.5	90.9	97.5	102.5	105.6	96.9	88.3	79.2	71.6
		MCWB	58.0	58.4	64.0	66.4	74.8	77.7	77.4	78.3	74.2	71.2	65.0	62.2
	2%	DB	64.4	70.0	76.8	81.3	88.1	93.3	99.4	99.8	91.5	83.5	72.8	64.2
		MCWB	55.4	58.3	62.5	65.6	73.5	76.5	78.2	77.9	73.9	70.2	60.7	56.8
	5%	DB	61.1	64.1	72.4	78.8	84.2	90.9	96.9	97.0	90.0	80.6	69.8	60.9
		MCWB	51.3	55.0	60.0	63.8	71.9	76.0	77.8	76.9	73.1	66.5	60.2	53.1
	10%	DB	54.9	60.8	68.5	74.5	81.4	89.6	93.1	91.6	86.3	75.1	64.5	55.3
		MCWB	46.5	52.5	58.0	62.5	69.8	75.7	76.8	75.8	71.4	65.0	56.3	48.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.5	63.6	67.2	71.9	78.1	81.0	83.2	82.4	78.2	74.4	67.5	63.4
		MCDB	66.7	70.7	77.8	81.0	87.9	91.8	95.4	95.9	88.5	84.1	75.5	68.7
	2%	WB	58.6	60.3	65.0	68.6	75.7	79.2	81.1	80.0	76.1	71.9	64.5	59.5
		MCDB	63.9	66.5	73.8	77.4	83.9	90.8	93.3	94.4	87.2	79.6	69.5	63.3
	5%	WB	52.5	57.1	62.8	66.5	73.8	77.7	79.4	78.4	74.7	69.7	61.2	55.0
		MCDB	58.9	62.5	69.8	74.5	81.7	89.0	92.5	92.4	85.5	74.7	67.1	58.8
	10%	WB	47.8	52.9	59.9	64.4	71.7	76.3	78.0	77.3	73.3	67.3	58.5	50.9
		MCDB	54.8	59.7	66.6	71.5	79.3	86.3	90.9	90.4	83.2	73.3	64.5	54.4
Mean Daily Temperature Range	5% DB	MDBR	22.2	22.0	23.1	24.1	22.1	21.3	23.3	23.3	24.0	24.5	24.4	21.1
		MCDBR	29.8	27.6	26.7	27.2	24.8	24.5	27.2	27.4	26.8	27.5	26.6	27.4
		MCWBR	19.9	18.0	15.4	14.9	12.2	10.5	9.4	9.1	10.3	13.7	16.6	19.0
	5% WB	MCDBR	25.4	23.8	22.6	21.0	21.3	22.3	23.6	24.2	22.6	19.6	22.9	22.0
		MCWBR	19.2	17.6	14.5	13.5	11.9	10.9	10.0	9.3	10.0	11.2	16.9	17.9
Clear-Sky Solar Irradiance	taub	0.305	0.310	0.350	0.385	0.424	0.427	0.445	0.446	0.393	0.349	0.317	0.297	
	taud	2.487	2.486	2.405	2.317	2.266	2.283	2.281	2.251	2.368	2.476	2.515	2.525	
	Ebn at Noon	285	296	292	285	274	272	266	264	273	278	278	281	
	Edh at Noon	26	30	35	40	43	42	42	43	36	30	25	24	
All-Sky Solar Radiation	RadAvg	797	965	1292	1616	1789	2043	2054	1864	1573	1175	855	679	
	RadStd	77	139	79	96	161	203	126	122	121	155	100	86	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page